QF620 Stochastic Modelling in Finance — Session 8 Reference Cheatsheet

Static Replication of European Payoffs: stochastic-vol parameters (ρ, ν), Leibniz's rule, the model-free framework, Breeden–Litzenberger, Carr–Madan replication, log contracts, variance swaps and VIX, and payoffs with discontinuities

Built from Session 8 slides (Tee Chyng Wen, SMU) · October 2026 · with symbol-by-symbol equation breakdowns

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Appendix B. Flag summary **10**

Source: "Session 8: Static Replication of European Payoffs" (Tee Chyng Wen, SMU, 29 slides). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged *[Beyond slides]*. ⚠marks a slide ambiguity or error, with the corrected version.

How to read the equations: every displayed equation is followed by a shaded **where** box that names each symbol and component and says what it represents in that formula.

Exam context: the exam format is not stated on the slides; check the course outline. From this session you need: how ρ and ν shape the return distribution and the smile; Leibniz's rule; the Breeden–Litzenberger derivation $\partial^2 C/\partial K^2 = e^{-rT} f(K)$; the Carr–Madan static replication formula, its derivation by parts, and how to apply it (aS_T^b, log contract); the variance swap pricing argument and $1/K^2$ weights; and replicating a digital with a call spread.

Notation used throughout (slide conventions)

Symbol	Meaning	Type	Notes
$f(s)$	risk-neutral density of S_T	function	[S10, S14]
$C(K)$, $P(K)$	today's call/put price at strike K, expiry T	price	[S10, S15]
$h(S_T)$	a European payoff; h', h'' its derivatives	function	[S16]
$F = S_0 e^{rT}$	forward price; the split point between puts and calls	price	[S16]
V_0	today's value of the payoff $h(S_T)$	price	[S16]
K	strike; inside integrals a dummy variable	number	[S16, S19]
ρ, ν	spot–vol correlation; vol-of-vol	number	[S2–S7]
σ_R^2, σ_K^2	realized variance; variance-swap strike	number	[S22]
σ_t	instantaneous (possibly random) volatility	random	[S23]
$\bar{K}$	digital strike	number	[S28]
$\mathbb{E}$, $\mathbb{E}^*$	risk-neutral expectation (slides use both)	operator	[S14, S23]

Notation trap: f is the risk-neutral density (not the real-world one) and F is the forward, not an antiderivative — except on [S8–S9], where F is an antiderivative in Leibniz's rule. K is both a strike and an integration variable. The slides write $\mathbb{E}$ and $\mathbb{E}^*$ interchangeably; all expectations in this session are under $\mathbb{Q}^*$.

0. Topic map

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PREREQUISITES
S6 smile, SABR (ρ, ν);   S7 martingale pricing V_0 = e^{−rT} E*[h(S_T)];   S4 Itô (d log S)
   |
   ▼
SESSION 8: STATIC REPLICATION OF EUROPEAN PAYOFFS
├─ STOCH-VOL PARAMETERS   [S2–S7]
|    ├─ ρ < 0: fat left tail, thin right tail ⇒ OTM puts up, OTM calls down      [S2–S4]
|    └─ ν > 0: kurtosis, both tails fat ⇒ OTM puts and calls up (curvature)     [S5–S7]
|
├─ LEIBNIZ'S RULE   [S8–S9]
|    └─ d/dx ∫_{u(x)}^{v(x)} f(x,t)dt = f(x,v)v' − f(x,u)u' + ∫ ∂f/∂x dt
|
├─ MODEL-FREE FRAMEWORK   [S10–S11]
|    └─ C(K) = e^{−rT}∫_K^∞ (s−K) f(s) ds; extract f from many quotes
|
├─ BREEDEN–LITZENBERGER (1978)   [S12–S15]
|    ├─ ∂C/∂K = −e^{−rT}∫_K^∞ f(s)ds                                          [S14]
|    └─ ∂²C/∂K² = ∂²P/∂K² = e^{−rT} f(K)                                      [S14–S15]
|
├─ CARR–MADAN (1998) STATIC REPLICATION   [S16–S21]
|    ├─ V_0 = ∫_0^F h ∂²P/∂K² dK + ∫_F^∞ h ∂²C/∂K² dK  (OTM puts and calls)    [S16]
|    ├─ Integrate by parts twice; put–call parity                             [S17–S18]
|    ├─ V_0 = e^{−rT}h(F) + ∫_0^F h''(K)P(K)dK + ∫_F^∞ h''(K)C(K)dK            [S19]
|    ├─ Example: aS_T^b, weights ab(b−1)K^{b−2}                               [S20]
|    └─ Log contract log(S_T/S_0): weights −1/K²                              [S21]
|
├─ VARIANCE SWAPS & VIX   [S22–S26]
|    ├─ Payoff N(σ_R² − σ_K²); σ_R² = (252/N)Σ log²(S_i/S_{i−1})              [S22]
|    ├─ σ²dt = 2dS/S − 2 d log S  ⇒ E∫σ²dt = 2rT − 2E log(S_T/S_0)           [S23–S24]
|    ├─ E*∫σ²dt = 2e^{rT}[∫_0^F P/K² dK + ∫_F^∞ C/K² dK]                      [S25]
|    └─ VIX: discretised version of the same strip                           [S26]
|
└─ DISCONTINUOUS PAYOFFS   [S27–S29]
     ├─ Start from ∫ h ∂²C/∂K² dK, integrate by parts                       [S27]
     └─ Digital call = −∂C(K̄)/∂K ≈ [C(K̄−ΔK) − C(K̄)]/ΔK (call spread)       [S28–S29]
   |
   ▼
LATER: smile-consistent pricing of CMS/exotic European payoffs; model-free implied variance
```

Dependency logic:

- Stochastic-vol parameters reshape the risk-neutral distribution (ρ skews it, ν fattens both tails). The smile is just the market's picture of that distribution.
- If calls trade at every strike, differentiating $C(K)$ twice in K recovers the density with no model (Breeden–Litzenberger). Leibniz's rule is the tool.
- Any smooth European payoff is an integral of the density, so substituting $f = e^{rT}\partial^2 C/\partial K^2$ and integrating by parts writes it as a bond + forward + a strip of OTM options (Carr–Madan).
- Itô turns realized variance into $2\frac{\mathrm{d}S}{S} - 2\,\mathrm{d}\log S$; the first part is a dynamic position worth $2rT$, the second is a log contract, which is replicated statically with $1/K^2$-weighted options. This is how variance swaps and VIX are priced.

1 Concept entries

1.1 Correlation ρ: distribution and pricing [S2–S4]

- ρ (spot–vol correlation) controls the **skewness** of returns [S2]. ⚠ The slide says "proportional"; the relation is monotone (higher ρ, more positive skew), not literally proportional.
- $\rho < 0$: when the stock falls, volatility rises, spreading out the left tail; the right tail has low vol and stays thin ⇒ **fat left tail, thin right tail** (negative skew) [S2–S3].
- Pricing: $\rho < 0$ raises OTM put prices and lowers OTM call prices relative to Black–Scholes; OTM puts gain most from a fat left tail [S2, S4].
- $\rho > 0$ has the opposite effect: fat right tail, thin left tail, positive skew [S2–S3].
- Smile picture [S4]: $\rho = -0.5$ gives implied vol falling with strike (downward skew); $\rho = 0$ is roughly symmetric; $\rho = +0.5$ tilts upward.

1.2 Vol-of-vol ν: distribution and pricing [S5–S7]

- $\nu = 0$: volatility is deterministic [S6], so log-returns are normal. ⚠ The slide says "(if $\beta = 0$)"; for normal *log*-returns the SABR backbone must be lognormal, $\beta = 1$. With $\beta = 0$ and $\nu = 0$ returns are normal in price units (Bachelier).
- $\nu > 0$ increases the **kurtosis** of returns: two fat tails [S5–S6]. This raises both OTM put and OTM call prices [S5, S7] — it adds curvature to the smile.
- If $\rho = 0$, raising ν only adds kurtosis: far-from-the-money options gain relative to near-the-money ones. Skewness comes from correlation [S5].
- Smile picture [S7]: larger ν lifts both wings (higher implied vol at low and high strikes).

1.3 Differentiating integrals; Leibniz's rule [S8–S9]

Chain rule for $g(u(x), v(x))$: $\frac{\mathrm{d}g}{\mathrm{d}x} = \frac{\partial g}{\partial u}u'(x) + \frac{\partial g}{\partial v}v'(x)$ [S8].

With F an antiderivative of f in t, $I(v,u,x) = \int_{u(x)}^{v(x)} f(x,t)\,\mathrm{d}t = F(x,v) - F(x,u)$, so $\partial I/\partial v = f(x,v)$, $\partial I/\partial u = -f(x,u)$. Hence

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_{u(x)}^{v(x)} f(x,t)\,\mathrm{d}t = f\big(x,v(x)\big)v'(x) - f\big(x,u(x)\big)u'(x) + \int_{u(x)}^{v(x)} \frac{\partial f(x,t)}{\partial x}\,\mathrm{d}t.$$

where

$f(x,v(x))v'(x)$	moving the upper limit adds a sliver of height f at the top.
$-f(x,u(x))u'(x)$	moving the lower limit removes a sliver at the bottom.
$\int \partial f/\partial x\,\mathrm{d}t$	the integrand itself changing with x (zero if $f = f(t)$ only).

Special case $f = f(t)$: $\frac{\mathrm{d}}{\mathrm{d}x}\int_{u(x)}^{v(x)} f(t)\,\mathrm{d}t = f(v(x))v'(x) - f(u(x))u'(x)$ [S9].

1.4 The model-free framework [S10–S11]

$$C(K) = e^{-rT}\mathbb{E}^*[(S_T - K)^+] = e^{-rT}\int_K^\infty (s-K)f(s)\,\mathrm{d}s.$$

where

$f(s)$	risk-neutral density of S_T: not specified by a model.
$\int_K^\infty$	payoff is zero for $s < K$.

- Model-based approach: choose dynamics, e.g. Black–Scholes, which gives

$$C(K) = \frac{e^{-rT}}{\sqrt{2\pi}} \int_{x^*}^{\infty} \left(S_0 e^{(r-\frac{\sigma^2}{2})T+\sigma\sqrt{T}x} - K\right) e^{-x^2/2} \mathrm{d}x$$

  with $x^* = \frac{\log(K/S_0)-(r-\sigma^2/2)T}{\sigma\sqrt{T}}$ *[Beyond slides]*. Bachelier, displaced diffusion or SABR also work. Once the model is calibrated, the density is fixed [S10].
- Model-free approach: with enough liquid quotes $C(K_1), C(K_2), \ldots$, extract f directly from prices [S11].

1.5 Implied risk-neutral density: motivation [S12–S13]

- Black–Scholes assumes $\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t^*$ under $\mathbb{Q}^*$ with one parameter σ [S12].
- Vanilla options are liquid, so prices are observed, not modelled; inverting gives one implied vol per strike [S12–S13].
- Next level: for a given maturity, find the **implied risk-neutral distribution** that matches the whole smile. The tool is Leibniz's rule [S13].

1.6 Breeden–Litzenberger (1978) [S14–S15]

Differentiate the call once and twice in K with Leibniz ($v = \infty$ does not depend on K, $u = K$, integrand $(s-K)f(s)$):

$$\frac{\partial C}{\partial K} = e^{-rT}\Big[\underbrace{0}_{\text{upper}} - \underbrace{(K-K)f(K)}_{=0} - \int_K^\infty f(s)\,\mathrm{d}s\Big] = -e^{-rT}\int_K^\infty f(s)\,\mathrm{d}s, \qquad \frac{\partial^2 C}{\partial K^2} = e^{-rT} f(K).$$

where

upper term	$\lim_{x\to\infty}(x-K)f(x)\frac{\mathrm{d}x}{\mathrm{d}K}$ on the slide: zero because the upper limit ∞ does not move with K.
$(K-K)f(K)$	lower-limit term: zero because the payoff vanishes at the strike.
$-\int_K^\infty f$	$\partial(s-K)/\partial K = -1$; so $-e^{rT}\partial C/\partial K = \mathbb{Q}^*(S_T > K)$ *[Beyond slides]*.
$e^{-rT}f(K)$	second derivative: only the lower-limit term survives $(-(-f(K)))$.

Puts: $P(K) = e^{-rT}\int_0^K (K-s)f(s)\,\mathrm{d}s$ gives $\frac{\partial P}{\partial K} = e^{-rT}\int_0^K f(s)\mathrm{d}s$ *[Beyond slides]* and

$$\boxed{\frac{\partial^2 C(K)}{\partial K^2} = \frac{\partial^2 P(K)}{\partial K^2} = e^{-rT} f(K)}$$

where

$f(K)$	the **implied distribution**: the density implicit in option prices, read off as the curvature of price in strike.

- Breeden–Litzenberger (1978): differentiate call or put prices twice in strike [S15]. *[Beyond slides]* A butterfly $[C(K-\Delta) - 2C(K) + C(K+\Delta)]/\Delta^2 \approx e^{-rT}f(K)$; convexity of C in K is a no-arbitrage condition ($f \geq 0$).
- Carr–Madan (1998): any European payoff can be replicated with cash, forwards, and European calls and puts [S15].

1.7 Static replication: setup [S16]

For a twice-differentiable payoff $h(S_T)$, with $F = S_0 e^{rT}$:

$$V_0 = e^{-rT}\mathbb{E}[h(S_T)] = e^{-rT}\int_0^\infty h(s)f(s)\,\mathrm{d}s = \underbrace{\int_0^F h(K)\frac{\partial^2 P(K)}{\partial K^2}\,\mathrm{d}K}_{(1)} + \underbrace{\int_F^\infty h(K)\frac{\partial^2 C(K)}{\partial K^2}\,\mathrm{d}K}_{(2)}.$$

where

$e^{-rT}f(K)$	replaced by $\partial^2 P/\partial K^2$ below F and $\partial^2 C/\partial K^2$ above (Breeden–Litzenberger); the discount factor is absorbed.
$s \to K$	dummy variable renamed to K as a reminder the derivatives are in strike.
split at F	uses liquid OTM/ATM options: low-strike puts, high-strike calls.

1.8 Integration by parts twice [S17–S18]

Call integral (2):

$$\int_F^\infty h\frac{\partial^2 C}{\partial K^2}\mathrm{d}K = \Big[h\frac{\partial C}{\partial K}\Big]_F^\infty - \Big[h'C\Big]_F^\infty + \int_F^\infty h''C\,\mathrm{d}K = -h(F)\frac{\partial C(F)}{\partial K} + h'(F)C(F) + \int_F^\infty h''(K)C(K)\,\mathrm{d}K.$$

where

∞ terms	$\partial C(\infty)/\partial K = 0$ and $C(\infty) = 0$ (assumed to beat the growth of h, h').

Put integral (1), same steps ($P(0) = 0$, $\partial P(0)/\partial K = 0$):

$$\int_0^F h\frac{\partial^2 P}{\partial K^2}\mathrm{d}K = h(F)\frac{\partial P(F)}{\partial K} - h'(F)P(F) + \int_0^F h''(K)P(K)\,\mathrm{d}K.$$

Combining [S18]:

$$V_0 = h(F)\Big[-\frac{\partial C(F)}{\partial K} + \frac{\partial P(F)}{\partial K}\Big] + h'(F)\big[C(F) - P(F)\big] + \int_0^F h''P\,\mathrm{d}K + \int_F^\infty h''C\,\mathrm{d}K.$$

1.9 Carr–Madan static replication formula [S18–S19]

Put–call parity $C(K) - P(K) = S_0 - Ke^{-rT}$ gives $\frac{\partial C}{\partial K} - \frac{\partial P}{\partial K} = -e^{-rT}$ and, at $K = F$, $C(F) - P(F) = S_0 - Fe^{-rT} = 0$. Hence

$$\boxed{V_0 = e^{-rT}h(F) + \int_0^F h''(K)P(K)\,\mathrm{d}K + \int_F^\infty h''(K)C(K)\,\mathrm{d}K}$$

where

$e^{-rT}h(F)$	a zero-coupon bond paying $h(F)$: the payoff's value at the forward.
(forward term)	$h'(F)[C(F) - P(F)]$: a position in $h'(F)$ forwards struck at F; worth zero today, so it vanishes from the price but is held in the hedge *[Beyond slides]*.
$h''(K)P(K)$	hold $h''(K)\,\mathrm{d}K$ OTM puts at each strike $K < F$.
$h''(K)C(K)$	hold $h''(K)\,\mathrm{d}K$ OTM calls at each strike $K > F$.
K	dummy variable: weights across a continuum of strikes.

[Beyond slides] Payoff-level identity behind it: $h(S_T) = h(F) + h'(F)(S_T - F) + \int_0^F h''(K)(K - S_T)^+\mathrm{d}K + \int_F^\infty h''(K)(S_T - K)^+\mathrm{d}K$. The hedge is *static*: set up once, never rebalanced. It is model-free: only today's option prices are needed.

1.10 Example: power payoff aS_T^b [S20]

$h = aS^b$, $h' = abS^{b-1}$, $h'' = ab(b-1)S^{b-2}$:

$$V_0 = e^{-rT}aF^b + \int_0^F ab(b-1)K^{b-2}P(K)\,\mathrm{d}K + \int_F^\infty ab(b-1)K^{b-2}C(K)\,\mathrm{d}K.$$

where

$a, b > 0$	real constants.
$ab(b-1)K^{b-2}$	option weight: positive for $b > 1$ (long options, convex payoff); negative for $0 < b < 1$ (short options); zero for $b = 1$ (just a forward).
check	*[Beyond slides]* under Black–Scholes $V_0 = e^{-rT}aS_0^b \exp\big[b(r - \frac{1}{2}\sigma^2)T + \frac{1}{2}b^2\sigma^2 T\big]$.

1.11 Log contract [S21] (solution *[Beyond slides]*)

Payoff $\log(S_T/S_0)$.

(1) **Black–Scholes:** $\log(S_T/S_0) = (r - \frac{1}{2}\sigma^2)T + \sigma W_T^*$, so

$$V_0 = e^{-rT}\Big(r - \frac{1}{2}\sigma^2\Big)T.$$

(2) **Carr–Madan:** $h(K) = \log(K/S_0)$, $h(F) = \log(F/S_0) = rT$, $h''(K) = -1/K^2$:

$$V_0 = e^{-rT}rT - \int_0^F \frac{P(K)}{K^2}\,\mathrm{d}K - \int_F^\infty \frac{C(K)}{K^2}\,\mathrm{d}K.$$

where

$-1/K^2$	short a strip of OTM options with weight $1/K^2$ (the log payoff is concave).
consistency	equating (1) and (2): $\int_0^F \frac{P}{K^2}\mathrm{d}K + \int_F^\infty \frac{C}{K^2}\mathrm{d}K = e^{-rT}\frac{\sigma^2 T}{2}$ under Black–Scholes.

1.12 Variance swaps: contract [S22]

$$\text{Payoff} = \text{Notional} \times (\sigma_R^2 - \sigma_K^2), \qquad \sigma_R^2 = \frac{252}{N}\sum_{i=1}^{N}\Big(\log\frac{S_{t_i}}{S_{t_{i-1}}}\Big)^2.$$

where

σ_R^2	realized (annualized) variance from daily log returns (zero mean assumed).
σ_K^2	strike variance; the **fair strike** makes the swap worth zero at inception.
252	trading days per year; N = days in the contract.

Gives pure volatility (variance) exposure without delta- or gamma-hedging vanilla options [S22].

1.13 Variance swaps: pricing [S23–S25]

Discrete sum $\approx$ integral: $\sum_i \log^2(S_{t_i}/S_{t_{i-1}}) \approx \int_0^T \sigma_t^2\mathrm{d}t$ [S23]. For $\mathrm{d}S_t = rS_t\mathrm{d}t + \sigma_t S_t \mathrm{d}W_t^*$, Itô gives

$$\mathrm{d}\log S_t = \frac{\mathrm{d}S_t}{S_t} - \frac{1}{2}\sigma_t^2\mathrm{d}t \implies \int_0^T \sigma_t^2\mathrm{d}t = 2\int_0^T \frac{\mathrm{d}S_t}{S_t} - 2\log\frac{S_T}{S_0}.$$

where

σ_t	any (stochastic) volatility: the argument is model-free in σ_t.
$2\int \mathrm{d}S/S$	continuously rebalanced position holding $2/S_t$ shares (dynamic).
$-2\log(S_T/S_0)$	short two log contracts (static).

Take $\mathbb{E}^*$. First term: $2\mathbb{E}^*\big[\int_0^T (r\,\mathrm{d}t + \sigma_t \mathrm{d}W_t^*)\big] = 2rT$. Second term, from the log contract (scale (2) of §1.11 by e^{rT}):

$$2\mathbb{E}^*\Big[\log\frac{S_T}{S_0}\Big] = 2\log\frac{F}{S_0} - 2e^{rT}\int_0^F \frac{P(K)}{K^2}\mathrm{d}K - 2e^{rT}\int_F^\infty \frac{C(K)}{K^2}\mathrm{d}K, \qquad \log\frac{F}{S_0} = rT.$$

Therefore [S25]

$$\boxed{\mathbb{E}^*\Big[\int_0^T \sigma_t^2\mathrm{d}t\Big] = 2e^{rT}\Big[\int_0^F \frac{P(K)}{K^2}\mathrm{d}K + \int_F^\infty \frac{C(K)}{K^2}\mathrm{d}K\Big], \qquad \sigma_K^2 = \frac{1}{T}\,\mathbb{E}^*\Big[\int_0^T \sigma_t^2\mathrm{d}t\Big]}$$

where

$2e^{rT}$	forward value (scale back from today's prices to time T).
$1/K^2$	weights: much more on low-strike puts than high-strike calls the swap is **skew-sensitive** [S25].
σ_K^2	*[Beyond slides]* fair annualized strike (divide total variance by T, matching the $252/N$ annualization).

⚠ [S25] writes $\mathbb{E}$ without the $*$; it is the risk-neutral expectation.

1.14 VIX index [S26]

$$\sigma^2 = \frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T}\left[\frac{F}{K_0} - 1\right]^2, \qquad \text{VIX} = 100\,\sigma.$$

where

$Q(K_i)$	mid of bid–ask of the OTM option at K_i: call if $K_i > K_0$, put if $K_i < K_0$, average of both at K_0.
K_0	first strike below the forward F (the discrete split point).
ΔK_i	$(K_{i+1} - K_{i-1})/2$; one-sided at the ends.
R, T	risk-free rate and time to expiry.
first term	discrete Riemann sum of $\frac{2e^{rT}}{T}\left[\int_0^F \frac{P}{K^2}\mathrm{d}K + \int_F^\infty \frac{C}{K^2}\mathrm{d}K\right]$ from §1.13.
correction	*[Beyond slides]* adjusts for splitting at $K_0 \neq F$ (the ITM sliver between K_0 and F); zero if $K_0 = F$.

1.15 Payoffs with discontinuities [S27–S28]

If h is not twice differentiable, do not plug into Carr–Madan directly. Start from

$$V_0 = e^{-rT}\int_0^\infty h(K) f(K)\,\mathrm{d}K = \int_0^F h(K)\frac{\partial^2 P}{\partial K^2}\mathrm{d}K + \int_F^\infty h(K)\frac{\partial^2 C}{\partial K^2}\mathrm{d}K$$

and integrate by parts twice on each smooth piece [S27].

Cash-or-nothing digital call [S28]: $h(S_T) = \mathbb{1}_{\{S_T \geq \bar{K}\}}$ with $\bar{K} \geq F$. Since $h = 0$ on $[F, \bar{K})$ and $h = 1$, $h' = h'' = 0$ on $[\bar{K}, \infty)$:

$$V_0 = \int_{\bar{K}}^\infty \frac{\partial^2 C}{\partial K^2}\mathrm{d}K = \left[h\frac{\partial C}{\partial K}\right]_{\bar{K}}^\infty - \left[h'C\right]_{\bar{K}}^\infty + \int_{\bar{K}}^\infty h''C\,\mathrm{d}K = -\frac{\partial C(\bar{K})}{\partial K}.$$

where

$-\partial C/\partial K$	$= e^{-rT}\mathbb{Q}^*(S_T > \bar{K})$; under Black–Scholes $e^{-rT}\Phi(d_2)$ *[Beyond slides].*
$\bar{K} \geq F$	so only the call region is involved; for $\bar{K} < F$ use the put region and parity *[Beyond slides].*

1.16 Digital via call spread [S29]

$$-\frac{\partial C(\bar{K})}{\partial K} \approx \frac{C(\bar{K} - \Delta K) - C(\bar{K})}{\Delta K}.$$

where

$1/\Delta K$	long $1/\Delta K$ calls at $\bar{K} - \Delta K$, short $1/\Delta K$ calls at $\bar{K}$.
payoff	0 below $\bar{K} - \Delta K$, linear ramp to 1 at $\bar{K}$, then 1: converges to the digital as $\Delta K \to 0$ (slide: $\bar{K} = 50$, $\Delta K = 0.10, 0.05, 0.01$).

[Beyond slides] This one-sided spread pays at least the digital, so it super-replicates; in practice the spread size is limited by strike spacing and the huge notional $1/\Delta K$.

1.17 Implementation *[Beyond slides]* (tested)

```
import numpy as np
from scipy.stats import norm
from scipy.integrate import quad
S0, r, sigma, T = 100.0, 0.05, 0.2, 1.0
F, D = S0*np.exp(r*T), np.exp(-r*T)
def call(K):
    d1 = (np.log(S0/K) + (r + 0.5*sigma**2)*T) / (sigma*np.sqrt(T))
    return S0*norm.cdf(d1) - K*D*norm.cdf(d1 - sigma*np.sqrt(T))
def put(K):                                          # put-call parity
    return call(K) - S0 + K*D
# 1) Breeden-Litzenberger: f(K) = e^{rT} d2C/dK2 (central difference)
K, h = 110.0, 0.01
f_BL = (call(K + h) - 2*call(K) + call(K - h)) / h**2 / D
m, s = np.log(S0) + (r - 0.5*sigma**2)*T, sigma*np.sqrt(T)
```

```
print(round(f_BL, 6), round(norm.pdf((np.log(K) - m)/s)/(K*s), 6))  # 0.017192 0.017192
# 2) Carr-Madan: V0 = D h(F) + int_0^F h''P dK + int_F^inf h''C dK
def carr_madan(h, h2):
    return (D*h(F) + quad(lambda k: h2(k)*put(k), 1e-8, F, limit=200)[0]
                   + quad(lambda k: h2(k)*call(k), F, np.inf, limit=200)[0])
a, b = 1.0, 2.0                                  # payoff a S_T^b
print(round(carr_madan(lambda k: a*k**b, lambda k: a*b*(b-1)*k**(b-2)), 4),
      round(D*a*S0**b*np.exp(b*(r - 0.5*sigma**2)*T + 0.5*b**2*sigma**2*T), 4))
                                                 # 10941.7428 10941.7428
# 3) Log contract log(S_T/S0): BS value D (r - sigma^2/2) T
print(round(carr_madan(lambda k: np.log(k/S0), lambda k: -1/k**2), 6),
      round(D*(r - 0.5*sigma**2)*T, 6))          # 0.028537 0.028537
# 4) Variance swap fair strike (2 e^{rT}/T)[int P/K^2 + int C/K^2] = sigma^2 under BS
I = quad(lambda k: put(k)/k**2, 1e-8, F)[0] + quad(lambda k: call(k)/k**2, F, np.inf)[0]
print(round(2*np.exp(r*T)*I/T, 6))               # 0.04
# 5) Digital call 1{S_T >= Kb} = -dC/dK = D N(d2), via a call spread
Kb, dK = 110.0, 0.01
d2 = (np.log(S0/Kb) + (r - 0.5*sigma**2)*T) / (sigma*np.sqrt(T))
print(round((call(Kb - dK) - call(Kb))/dK, 4), round(D*norm.cdf(d2), 4))  # 0.3539 0.3539
```

2 Comparison tables

2.1 What ρ and ν do [S2–S7]

	$\rho < 0$	$\rho > 0$	$\nu \uparrow$ ($\rho = 0$)
Distribution	fat left, thin right tail (neg. skew)	fat right, thin left tail (pos. skew)	both tails fat (kurtosis)
OTM puts	up	down	up
OTM calls	down	up	up
Smile	downward skew	upward skew	curvature (both wings up)

2.2 Replication weights by payoff

Payoff $h(S_T)$	$h(F)$	$h''(K)$	**Replicating portfolio**
aS_T^b [S20]	aF^b	$ab(b-1)K^{b-2}$	bond aF^b, abF^{b-1} forwards, options weighted $ab(b-1)K^{b-2}$
$\log(S_T/S_0)$ [S21]	rT	$-1/K^2$	bond rT, $1/F$ forwards, short OTM options weighted $1/K^2$
S_T^2 *[Beyond slides]*	F^2	2	bond F^2, $2F$ forwards, 2 of every OTM option
$\mathbb{1}_{\{S_T \geq \bar{K}\}}$ [S28]	—	not smooth	limit of call spreads: $-\partial C(\bar{K})/\partial K$

2.3 Static vs dynamic replication

	Static (Carr–Madan) [S15–S19]	**Dynamic (Black–Scholes delta hedge)**
Instruments	bond, forward, strip of OTM options	stock and bond
Rebalancing	none: set up once	continuous
Model needed	none (needs option prices at all strikes)	yes (dynamics and σ)
Payoffs	European, depend on S_T only	any replicable claim

3 Master formula sheet

Each line repeats the key symbols so the sheet can be used on its own.

Leibniz [S9]

- $\frac{\mathrm{d}}{\mathrm{d}x}\int_{u(x)}^{v(x)} f(x,t)\mathrm{d}t = f(x,v)v' - f(x,u)u' + \int_u^v \partial_x f \,\mathrm{d}t.$

Breeden–Litzenberger [S14–S15]

- $\frac{\partial C}{\partial K} = -e^{-rT}\int_K^\infty f(s)\mathrm{d}s$; $\frac{\partial^2 C}{\partial K^2} = \frac{\partial^2 P}{\partial K^2} = e^{-rT}f(K)$.
  where f = risk-neutral density of S_T.

Carr–Madan [S19]

- $V_0 = e^{-rT}h(F) + \int_0^F h''(K)P(K)\mathrm{d}K + \int_F^\infty h''(K)C(K)\mathrm{d}K$, $F = S_0e^{rT}$.
  uses $\partial_K C - \partial_K P = -e^{-rT}$, $C(F) = P(F)$.
- Power: $h'' = ab(b-1)K^{b-2}$. Log contract: $h(F) = rT$, $h'' = -1/K^2$.

Variance swap [S22–S25]

- $\sigma_R^2 = \frac{252}{N}\sum \log^2 \frac{S_{t_i}}{S_{t_{i-1}}}$; $\int_0^T \sigma_t^2\mathrm{d}t = 2\int_0^T \frac{\mathrm{d}S}{S} - 2\log\frac{S_T}{S_0}$.
- $\mathbb{E}^*\int_0^T \sigma_t^2\mathrm{d}t = 2e^{rT}\big[\int_0^F \frac{P}{K^2}\mathrm{d}K + \int_F^\infty \frac{C}{K^2}\mathrm{d}K\big]$; $\sigma_K^2 = \frac{1}{T}\times$ that.
- VIX: $\sigma^2 = \frac{2}{T}\sum \frac{\Delta K_i}{K_i^2}e^{RT}Q(K_i) - \frac{1}{T}(\frac{F}{K_0}-1)^2$, VIX $= 100\sigma$.

Digital [S28–S29]

- $V_0 = -\partial C(\bar K)/\partial K \approx [C(\bar K - \Delta K) - C(\bar K)]/\Delta K$.

4 Glossary

- **Breeden–Litzenberger formula:** $\partial^2 C/\partial K^2 = e^{-rT}f(K)$ [S15].
- **Butterfly spread:** discrete second difference in strike; approximates the density *[Beyond slides]*.
- **Call spread:** long low-strike, short high-strike call; approximates a digital [S29].
- **Carr–Madan formula:** static replication of a European payoff with bond, forwards and OTM options [S15, S19].
- **Cash-or-nothing digital:** pays 1 if $S_T \geq \bar K$ [S28].
- **Implied (risk-neutral) distribution:** the density implicit in option prices [S13, S15].
- **Kurtosis:** tail fatness; raised by vol-of-vol [S5–S6].
- **Leibniz's rule:** differentiating an integral with variable limits [S9].
- **Log contract:** pays $\log(S_T/S_0)$ [S21].
- **Model-free:** prices from market option quotes, no dynamics assumed [S10].
- **Skewness:** asymmetry of returns; driven by ρ [S2].
- **Static replication:** hedge set up once and held to expiry [S16].
- **Variance swap:** pays notional $\times(\sigma_R^2 - \sigma_K^2)$ [S22].
- **VIX:** CBOE index; 100× square root of 30-day model-free implied variance [S26].

5 Self-test

Questions

Q1 ([S2–S7]). How do $\rho < 0$ and a larger ν change the return distribution and OTM option prices?
Q2 ([S9]). State Leibniz's rule.
Q3 ([S14]). Derive $\partial C/\partial K$ and $\partial^2 C/\partial K^2$ from $C(K) = e^{-rT}\int_K^\infty (s-K)f(s)\mathrm{d}s$.
Q4 ([S16–S19]). Derive the Carr–Madan formula. Where are put–call parity and $C(F) = P(F)$ used?
Q5 ([S20]). Replicate aS_T^b. When are the options held short?
Q6 ([S21]). Value the log contract under Black–Scholes and by static replication.
Q7 ([S23]). Show $\int_0^T \sigma_t^2\mathrm{d}t = 2\int \mathrm{d}S/S - 2\log(S_T/S_0)$.
Q8 ([S24–S25]). Derive the fair variance and explain the $1/K^2$ weights.
Q9 ([S28–S29]). Value a digital call with $\bar K \geq F$ and build a finite replication.

Solutions

S1. $\rho < 0$: fat left tail, OTM puts up, OTM calls down (skew). $\nu \uparrow$: fatter tails both sides, OTM puts and calls up (curvature).
S2. See §1.3.
S3. $-e^{-rT}\int_K^\infty f\mathrm{d}s$; $e^{-rT}f(K)$ (see §1.6).

S4. See §1.8–1.9: parity gives $-\partial_K C(F) + \partial_K P(F) = e^{-rT}$ and kills the $h'(F)$ term.
S5. §1.10; short when $0 < b < 1$ (concave payoff).
S6. $e^{-rT}(r - \frac{1}{2}\sigma^2)T$; $e^{-rT}rT - \int_0^F \frac{P}{K^2} - \int_F^\infty \frac{C}{K^2}$.
S7. Itô on $\log S$: $\mathrm{d}\log S = \mathrm{d}S/S - \frac{1}{2}\sigma^2\mathrm{d}t$; rearrange and integrate.
S8. See §1.13; $1/K^2$ puts more weight on low strikes, so the price depends on the skew.
S9. $-\partial C(\bar{K})/\partial K = e^{-rT}\mathbb{Q}^*(S_T > \bar{K})$; long and short $1/\Delta K$ calls at $\bar{K} - \Delta K$ and $\bar{K}$.

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title	header
S2–S4	ρ: distribution, pricing, smile	1.1
S5–S7	ν: distribution, pricing, smile	1.2
S8–S9	Differentiation of integrals; Leibniz's rule	1.3
S10–S11	Model-free framework	1.4
S12–S13	Implied risk-neutral density: motivation	1.5
S14–S15	Breeden–Litzenberger; Carr–Madan statement	1.6
S16	Static replication setup	1.7
S17–S18	Integration by parts; combining	1.8
S19	Carr–Madan formula	1.9
S20	Example aS_T^b	1.10
S21	Log contract (question)	1.11
S22	Variance swap contract	1.12
S23–S25	Variance swap pricing	1.13
S26	VIX formula	1.14
S27	Discontinuous payoffs	1.15
S28	Cash-or-nothing digital	1.15
S29	Call spread implementation	1.16

Appendix B. Flag summary (⚠)

1. [S5] "Normal log-return distribution (if $\beta = 0$)" should read $\beta = 1$; $\beta = 0$ with $\nu = 0$ is the normal (Bachelier) model in price units.
2. [S2] ρ is said to be "proportional" to skewness; the relationship is monotone, not proportional.
3. [S25] The final variance formula uses $\mathbb{E}$; it is the risk-neutral $\mathbb{E}^*$. The fair annualized strike divides it by T *[Beyond slides]*.
4. [S8–S9] F denotes an antiderivative there, but the forward S_0e^{rT} from [S16] on.
5. [S21] The log-contract question has no worked answer on the slides; it is answered here with my own working.